Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Driver 1 and matrix

File: matrix1.kicad_sch

Driver 2 and matrix

File: matrix2.kicad_sch

Parallelogram 4-layer density study (p2-132): 132 XL-1615RGBC-RF LEDs (6x22 skewed grid, 3.70x2.60 mm), 2x AW20198QNR QFN-40, 4-layer JLCPCB standard.
25 x 70 mm tip-to-tip skewed parallelogram, 20 x 35 mm prismatic LiPo cell.
Solid GND thermal pads on U3, U5, and all AW20198QNR QFN-40 drivers (0 via-in-pad).

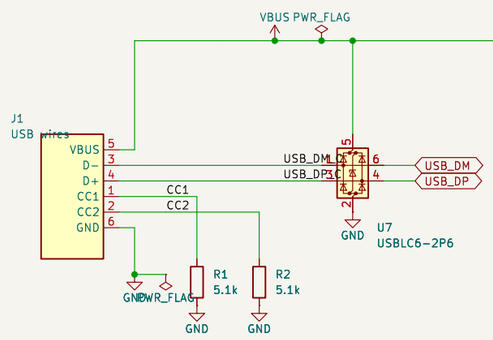
Drawn by scripts/parallelogram/sch_gen.py
132 XL-1615RGBC-RF LEDs (6x22 skewed grid, 3.70x2.60 mm), 2x AW20198QNR QFN-40, 4-layer JL
**CERN-OHL-P-2.0**

Sheet: /
File: parallelogram.kicad_sch

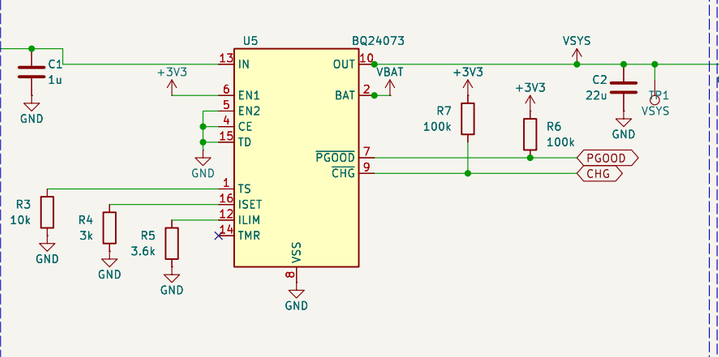
***Title: parallelogram (p2-132): top***

| Size: A4 | Date: 2026-11-05 | **Rev: P1** |
|---|---|---|
| KiCad E.D.A. 9.0.8 | | Id: 1/5 |

## USB WIRES, ESD

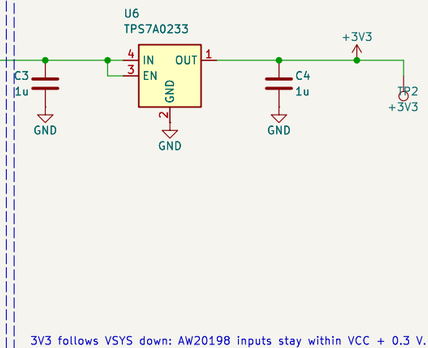



## CHARGER AND POWER PATH




USB500: 450–500 mA in, 297 mA charge, ILIM must not float.
TMR open: default 5 h. TS fixed: no NTC.

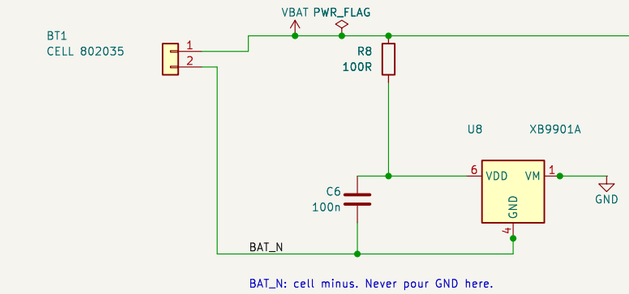
## 3V3




3V3 follows VSYS down: AW20198 inputs stay within VCC + 0.3 V.
150 mA max.

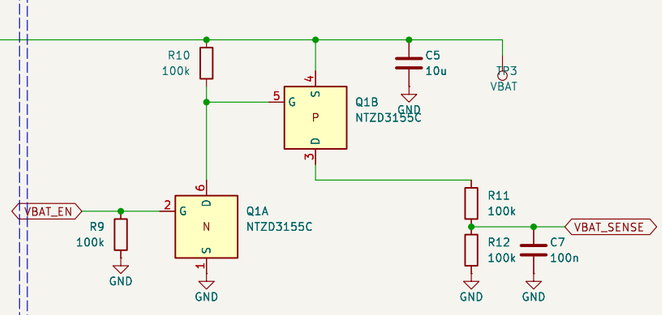
## CELL AND PROTECTION

Hand−soldered leads. Standard 802035 / 102035 (20x35 mm) prismatic LiPo cell.




BAT_N: cell minus. Never pour GND here.

## VBAT SENSE



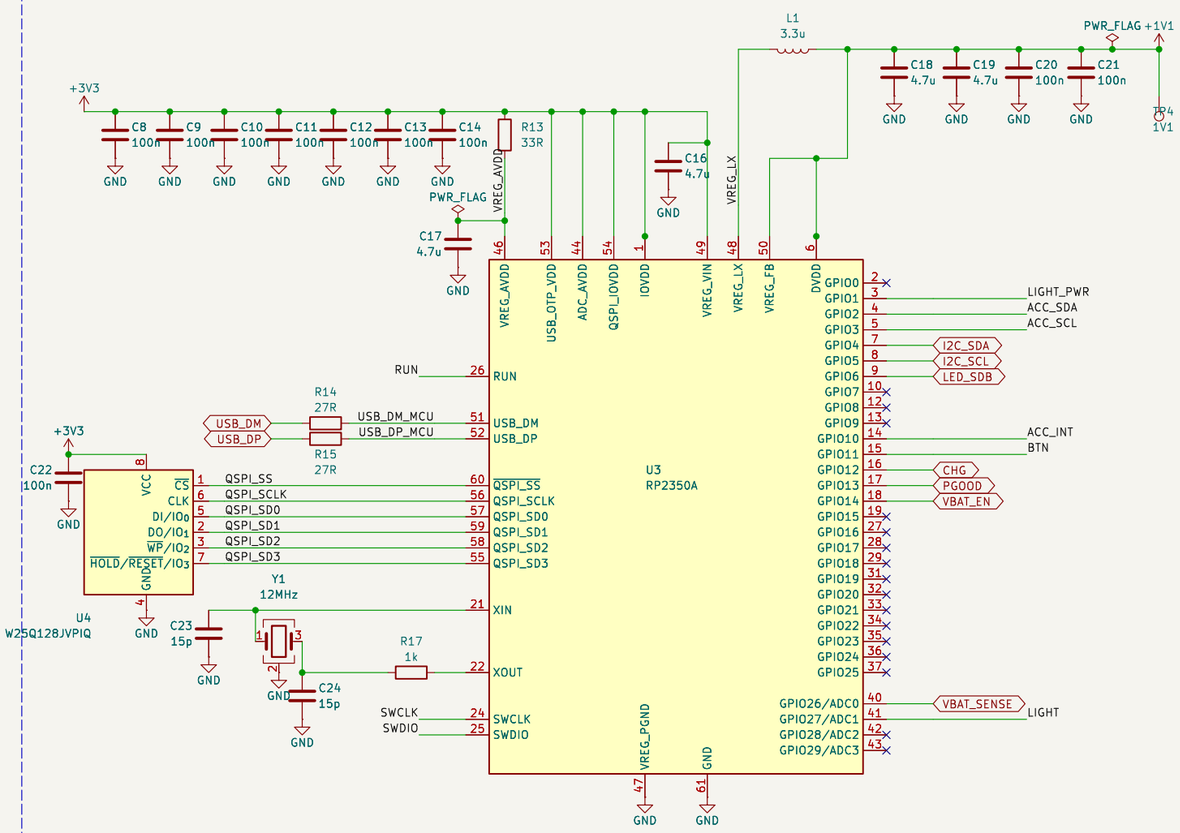

Drawn by scripts/parallelogram/sch_gen.py
USB−C wires, charger, protection, LDO, VBAT sense (prismatic 802035 cell)
**CERN−OHL−P−2.0**

Sheet: /Power/
File: power.kicad_sch

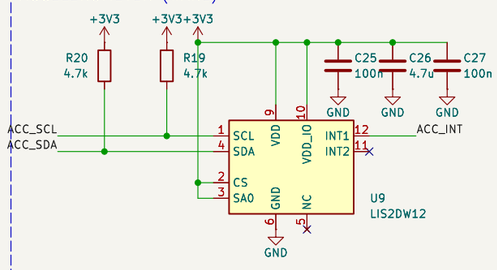
***Title: parallelogram (p2−132): power***

| Size: A3 | Date: 2026−11−05 | **Rev: P1** |
|---|---|---|
| KiCad E.D.A. 9.0.8 | | Id: 2/5 |

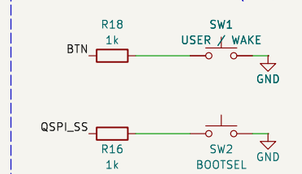
## RP2350A, FLASH, CRYSTAL



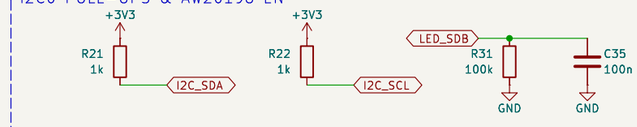

## ACCELEROMETER (WAKE)



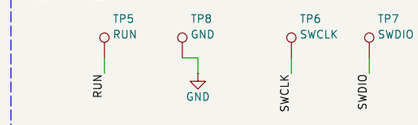

## PEG-LESS SIDE BUTTONS (EVQP7J01P)



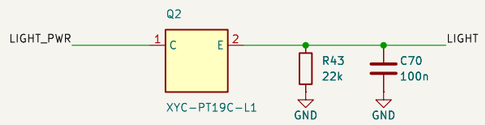

## I2C0 PULL-UPS & AW20198 EN




## TEST PADS




## AMBIENT LIGHT SENSOR (TOP BEZEL)



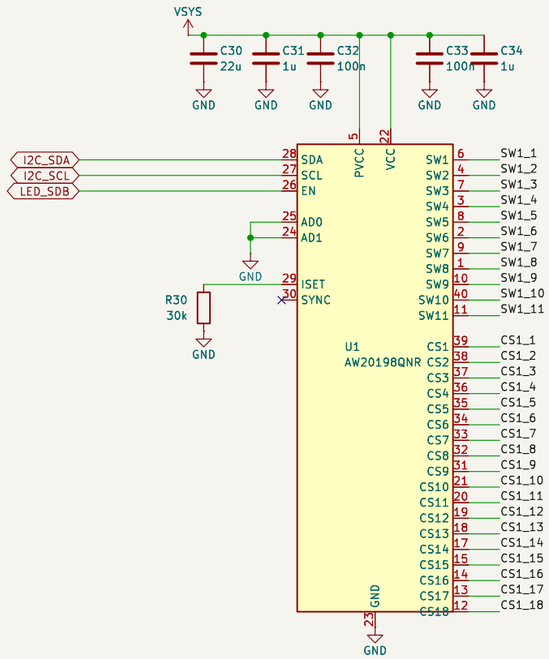

Drawn by scripts/parallelogram/sch_gen.py
RP2350A, flash, crystal, accelerometer, peg-less side buttons, top-bezel light sensor
**CERN-OHL-P-2.0**

Sheet: /MCU/
File: mcu.kicad_sch

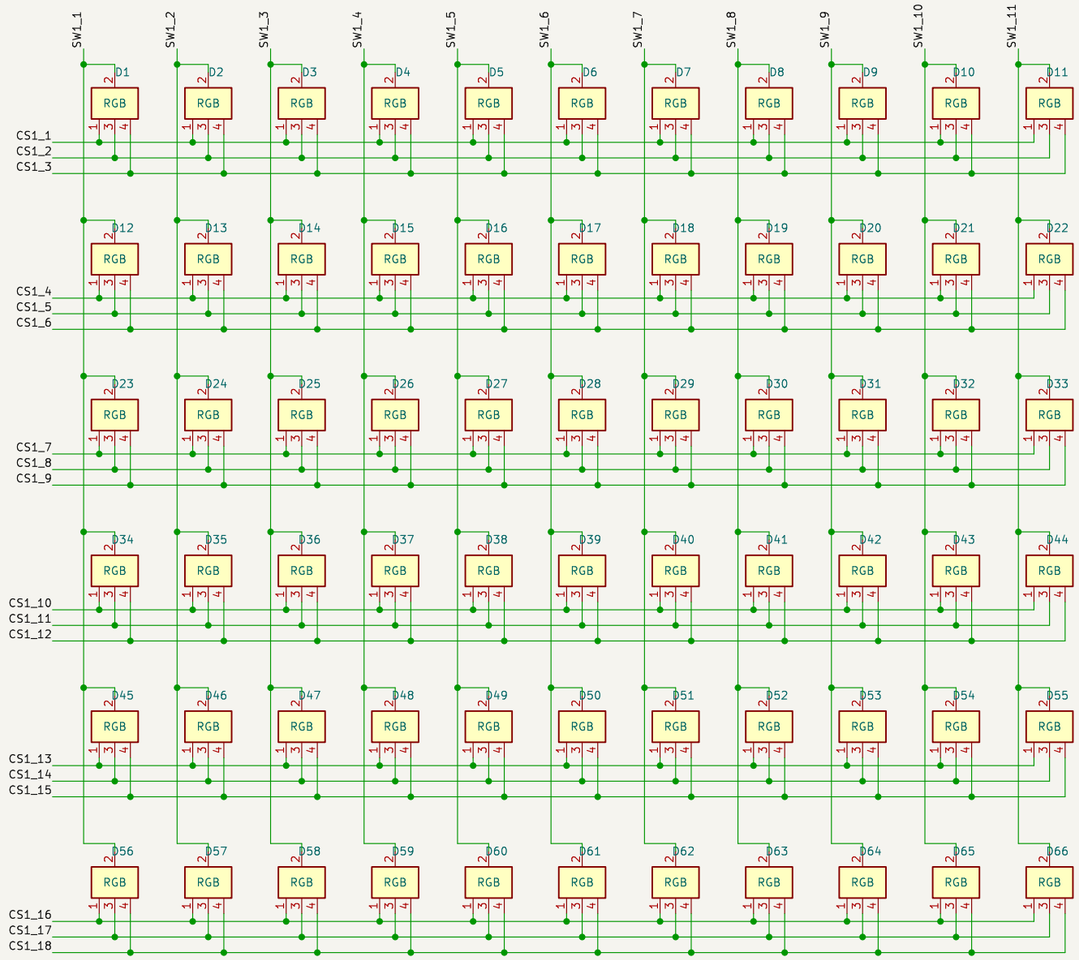
***Title: parallelogram (p2-132): mcu***

| Size: A3 | Date: 2026-11-05 | **Rev: P1** |
|---|---|---|
| KiCad E.D.A. 9.0.8 | | Id: 3/5 |

## DRIVER 1 (U1, 0x20)




## 66 RGB LEDS: R=CS(3k−2), G=CS(3k−1), B=CS(3k)



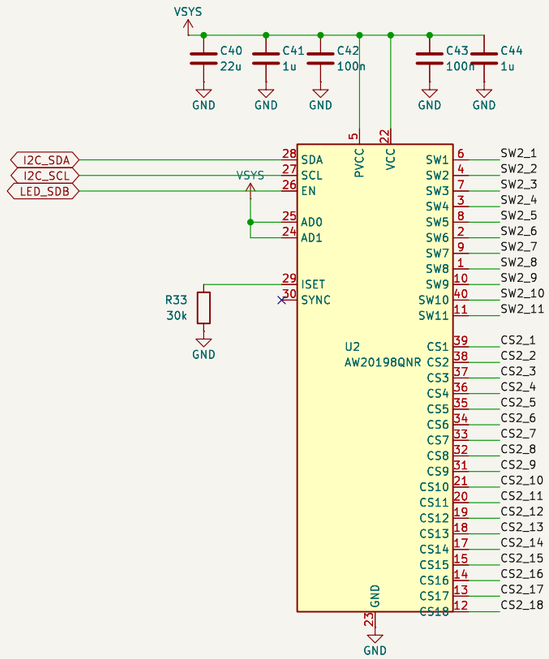

Drawn by scripts/parallelogram/sch_gen.py
U1 AW20198QNR (0x20), 66 RGB LEDs (drivers-aw20198.md)
**CERN-OHL-P-2.0**

Sheet: /Driver 1 and matrix/
File: matrix1.kicad_sch

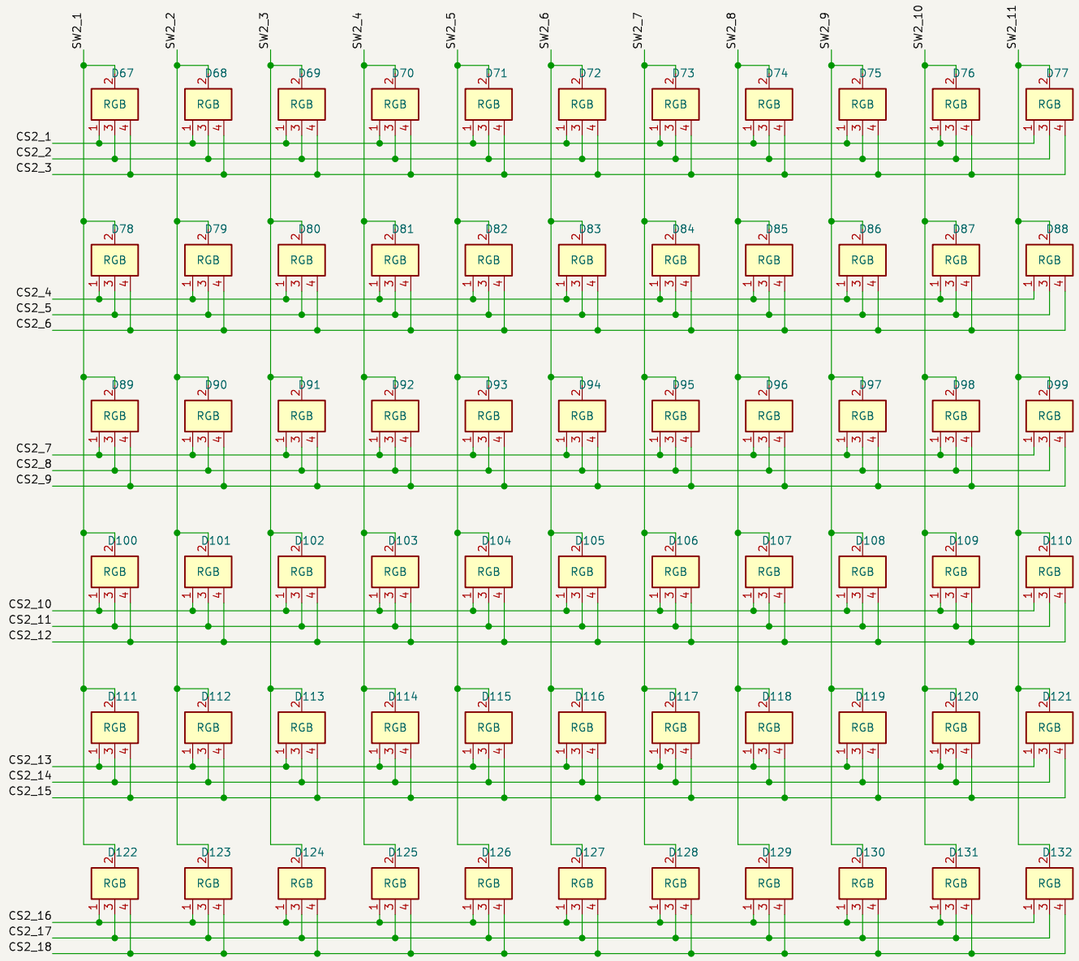
***Title: parallelogram (p2-132): driver 1 and matrix***

| Size: A3 | Date: 2026-11-05 | **Rev: P1** |
|---|---|---|
| KiCad E.D.A. 9.0.8 | | Id: 4/5 |

## DRIVER 2 (U2, 0x25)




## 66 RGB LEDS: R=CS(3k−2), G=CS(3k−1), B=CS(3k)




Drawn by scripts/parallelogram/sch_gen.py
U2 AW20198QNR (0x25), 66 RGB LEDs (drivers-aw20198.md)
**CERN-OHL-P-2.0**

Sheet: /Driver 2 and matrix/
File: matrix2.kicad_sch

***Title: parallelogram (p2-132): driver 2 and matrix***

| Size: A3 | Date: 2026-11-05 | **Rev: P1** |
|---|---|---|
| KiCad E.D.A. 9.0.8 | | Id: 5/5 |